

Hardy-Weinberg example



HARDY-WEINBERG FORMULA

$$p^2 + 2pq + q^2 = 1.0$$

p = % of DOMINANT allele (B)

q = % of RECESSIVE allele (b)

p^2 = % of the HOMOZYGOUS DOMINANT (BB)

$2pq$ = % of the HETEROZYGOTES (Bb)

q^2 = % of the HOMOZYGOUS RECESSIVE (bb)

6 students total

BLUE eyes: 2 out of 6 (0.33 or 33%)

BROWN eyes: 4 out of 6 (0.67 or 67%)

If you know the % of people that have the HOMOZYGOUS RECESSIVE condition (**BLUE** eyes), you can calculate the rest of the values.

$$q^2 = 2/6 = 0.33 \text{ or } 33.0\% \text{ (frequency of HOMOZYGOUS RECESSIVE, bb BLUE eyes)}$$

$$q = 0.574 \text{ or } 57.4\% \text{ (frequency of BLUE allele). We know this by taking the square root of } q^2$$

$$p = 1.0 - q = 1.0 - 0.574 = 0.426 \text{ or } 42.6\% \text{ (frequency of BROWN allele)}$$

$$p^2 = p \times p = 0.426 \times 0.426 = 0.181 \text{ or } 18.1\% \text{ (frequency of HOMOZYGOUS DOMINANT, BB BROWN eyes)}$$

$$2pq = 2 \times 0.426 \times 0.574 = 0.489 \text{ or } 48.9\% \text{ (frequency of HETEROZYGOTES, Bb BROWN eyes)}$$

Quick check >>>>>>>>> $p^2 + 2pq + q^2$ should equal 1.0 or 100.0%

$$0.181 + 0.489 + 0.330 = 1.0 \text{ >>>>>>>> SOLVED!}$$